



**Revised
Generation Interconnection
System Impact Study Report
for
Queue Project AE1-160
VENANGO 34.5 KV
12.7 MW Capacity / 20 MW Energy**

January, 2020

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1 Introduction

This System Impact Study (SIS) has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the Feasibility Study Agreement between Titusville Solar, LLC, the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Pennsylvania Electric Company (PENELEC).

2 Revision from August 2019 System Impact Study Report

The System Impact Study report has been revised to include results from the Stability Analysis and Reactive Power Assessment in Section 19.

3 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

4 General

The Interconnection Customer (IC) has proposed a Solar generating facility located in Venango County, Pennsylvania. The installed facilities will have a total capability of 20 MW with 12.7 MW of this output being recognized by PJM as Capacity. The proposed Commercial Operation Date for this project is on or before December 31, 2022.

Queue Number	AE1-160
Project Name	VENANGO 34.5 KV
Interconnection Customer	Titusville Solar, LLC
State	Pennsylvania
County	Venango
Transmission Owner	PENELEC
MFO	20
MWE	20
MWC	12.7
Fuel	Solar
Basecase Study Year	2022

4.1 Point of Interconnection

AE1-160 will interconnect with the PENELEC distribution system via a tap on the 34.5 kV Colonel Drake circuit at pole # T1-35751. The IC's proposed generating unit site is approximately 3.8 miles south of Titusville, PA., off William Flinn Highway.

Attachment 1 shows a one-line diagram of the proposed primary direct connection facilities for the AE1-160 generation project to connect to the PENELEC distribution system. Attachment 2 provides the proposed location for the point of interconnection. IC will be responsible for constructing all of the facilities on its side of the POI, including the attachment facilities which connect the generator to the PENELEC distribution system's direct connection facilities.

4.2 Cost Summary

The AE1-160 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$176,200
Direct Connection Network Upgrade	\$0
Non Direct Connection Network Upgrades	\$14,500
New System Upgrades	\$0
Contribution to Previously Identified Upgrades	\$0
Total Costs	\$190,700

The costs provided above exclude the Contribution in Aid of Construction ("CIAC") Federal Income Tax Gross Up charge. If, at a future date, it is determined that the CIAC Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

The required Attachment Facilities and Direct and Non-Direct Connection work for the interconnection of the AE1-160 generation project to the PENELEC Distribution System is detailed in the following sections. The associated one-line is shown in Attachment 1.

5 Transmission Owner Scope of Work

The AE1-160 project will interconnect with the PENELEC distribution system via a tap on the 34.5 kV Colonel Drake circuit at pole # T1-35751. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct the new interconnection station and the associated facilities.

Attachment 1 shows a one-line diagram of the proposed connection facilities for the AE1-160 generation project to connect to the PENELEC distribution system. Attachment 2 provides the proposed location for the point of interconnection. The IC will be responsible for constructing all of the facilities on its side of the POI, including the attachment facilities which connect the generator to the FE distribution system's connection facilities.

5.1 Attachment Facilities

The total preliminary cost estimate for the Attachment work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Tap on the Colonel Drake 34.5 kV line and associated 34.5 kV metering package.	\$176,200
Total Attachment Facility Costs	\$176,200

5.2 Direct Connection Cost Estimate

There is no Direct Connection scope of work required.

5.3 Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Titusville 34.5kV SS. Remote Relay and Metering Settings at Titusville 34.5kV.	\$14,500
Total Non-Direct Connection Facility Costs	\$14,500

6 Schedule

Based on the scope of work, it is expected to take a minimum of **6 months** after the signing of a Wholesale Market Participation Agreement and construction kickoff call to complete the installation. This assumes that there will be no environmental issues, that there will be no delays in acquiring any necessary permits for implementing the defined work, and that any distribution system outages will be allowed when requested.

7 Transmission Owner Analysis

PENELEC performed an analysis of its distribution system. The AE1-160 project did not contribute to any overloads on the distribution system.

8 Interconnection Customer Requirements

8.1 System Protection

An analysis was conducted to assess the impact of the Venango 34.5 kV (AE1-160) Project on the system protection requirements in the area. The results of this review show that the following relay additions will be required:

Proposed single line diagrams show the IC constructing a generation facility they call “Venango” tapping Penelec’s Titusville-34.5kV Colonel Drake at pole T1-35751.

The 34.5kV interconnection proposal will require Developer to meet applicable "Technical Requirements" as outlined in First Energy's document titled “Technical Requirements for the Interconnection of Customer-Owned Generation to the FirstEnergy Distribution System”.

Protection requirements are included in the "Technical Requirements" document.

8.2 General Concerns

It is to be understood, for abnormal operation of the Penelec system, which could cause Developer’s generation facility to be electrically isolated from the Penelec system synchronous source via the tripping of a interconnecting primary voltage line or device, the IC will, via Penelec’s direction, be required to disconnect the generation from Penelec’s system and remain disconnected (**units are required to be OFF LINE**), until the Penelec system normal circuitry is restored. These abnormal conditions will be reviewed by Penelec system operators as to the need for the generation facility to be disconnected.

8.3 Requirements for Owner’s/Developer’s generation IPP Facility

The proposed interconnection Customer’s facilities must be designed in accordance with the document titled *FirstEnergy Distribution Engineering Practices Interconnection of Customer-Owned Generation to the FirstEnergy Distribution System* dated 11/17/14 located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>

The document is referred to as engineering practice EP(# 02-280) with section 4 part C specifically referencing the “interconnection technical requirements”. Certain protection requirements are shown.

Additionally, Owner/Developer is responsible to provide adequate protection (for their equipment) under any distribution system operating condition' - which includes 'Separation from supply' (i.e. tripping of F.E. circuit breakers) and 'Re-synchronizing the generation after electric restoration of the supply' (i.e. reclosing of F.E. circuit breakers).

IC's protection must be designed to coordinate with the reclosing practices of FirstEnergy line protective devices. The generator must cease to energize the FirstEnergy circuit to which it is connected prior to reclosing of any (FE) automatic reclosing devices.

IC's electrical protection and control schematics shall be provided to FE for consideration. FE may request modifications, if required, to meet the technical requirements.

8.4 Power Factor Requirements

IC will be responsible for meeting a power factor between 0.95 lagging (producing MVARs) to 0.95 leading (absorbing MVARs) and assure that voltage deviation will be less than 1.0 volt as measured at the POI under all Solar Gen operating conditions due to the inherent dynamic reactive power capability of this solar facility.

Generators with no inherent VAR (reactive power) control capability, or those that have a restricted VAR capability less than the defined requirements, must provide dynamic supplementary reactive support located at the generation facility with electrical characteristics equivalent to that provided by a similar sized synchronous generator. A Dynamic Reactive Compensation (either Static VAR Compensator (SVC) or STATCOM) or other method be applied in order to maintain the required specifications at the POI. IC is responsible for the installation of equipment on its side of the POI in order to adhere to the criteria stated above by FirstEnergy.

9 Revenue Metering and SCADA Requirements

9.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

9.1.1 Meteorological Data Reporting Requirement

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

9.2 PENELEC Requirements

IC will be required to comply with all FE Revenue Metering Requirements for Generation Interconnection Customers. These FE requirements are the following:

The FE operating company (Penelec) shall provide, own, operate, test, and maintain the revenue metering equipment at the Interconnection Customer's (IC) expense. The revenue metering equipment includes, but is not limited to, current transformers, voltage transformers, secondary wires, meter socket, bidirectional revenue meter, and associated devices. The IC shall mount the instrument transformers unless otherwise agreed to by Penelec. The instrument transformers and meter socket shall be installed in a location that is readily accessible to authorized Penelec representatives. Penelec will provide the IC access to bidirectional kWh and kVARh pulses from the Penelec meter at the IC's expense if requested. The IC shall, at its expense, install, own, operate, test, and maintain any metering and telemetry equipment that may be required to provide real-time meter data to FE or PJM.

10 Network Impacts

The Queue Project AE1-160 was evaluated as a 20.0 MW (Capacity 12.7 MW) injection at Venango 34.5kV in the PENELEC area. Project AE1-160 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AE1-160 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Load Flow

11 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

12 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

13 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

14 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
21856581	939385	AE1-169 TAP1	PENELEC	200571	26UNION CY	PENELEC	1	PN-P1-2-PN-345-001	operation	120.0	79.8	85.86	AC	7.27

15 System Reinforcements

None

Affected Systems

16 Affected Systems

16.1 NYISO

None

17 Contingency Descriptions

Contingency Name	Contingency Definition
PN-P1-2-PN-345-001	CONTINGENCY 'PN-P1-2-PN-345-001' /* ERIE WEST - WAYNE 345KV DISCONNECT BRANCH FROM BUS 200599 TO BUS 200595 CKT 1 /* 26ERIE W 345 26WAYNE 345 END

Short Circuit

18 Short Circuit

The following Breakers are overduty:

None

Stability and Reactive Power Assessment

19 Stability Analysis and Reactive Power Assessment

19.1 Executive Summary

Generator Interconnection Request AE1-160 is for a 20 MW Maximum Facility Output (MFO) Solar Facility. AE1-160 consists of 7 x 2.857 MW Power Electronics FS3350M inverters with a Point of Interconnection (POI) tapping the 34.5 kV line from Titusville Substation to Colonel Drake Substation in Venango County, Pennsylvania, in the First Energy (FE) transmission system.

The power flow scenario for the analysis was based on the RTEP 2022 summer peak case, modified to include applicable queue projects. AE1-160 has been dispatched online at maximum facility output, with approximately unity power factor at the high-side of the station transformer.

AE1-160 was tested for compliance with NERC, PJM, Transmission Owner and other applicable criteria. For this study, 54 contingencies were simulated, each with a 20 second simulation time period. Studied faults included:

- Steady-state operation (20 second simulation)
- Three-phase faults with normal clearing time
- Single-phase bus faults with normal clearing time
- Single-phase faults with a stuck breaker
- Single-phase faults placed at 80% of the line with delayed (Zone 2) clearing at remote line end because of primary communications/relaying failure

The 54 fault contingencies tested on the 2022 summer peak case met the recovery criteria:

The AE1-160 generator was able to ride through the faults except for faults where protective actions trip one or more generator(s).

All generators maintained synchronism and any post-contingency oscillations are positively damped with a damping margin of at least 3%.

All bus voltages recover to 0.7 p.u. within 2.5 seconds and the final voltage is within the range of 0.92 p.u. to 1.05 p.u. for buses other than 500 kV. The final voltages for 500 kV buses should be within 1.02 p.u. to 1.08 p.u.

No transmission element trips, other than those either directly connected or designated to trip as a consequence of the fault.

AE1-160 meets both the 0.95 lagging and 0.95 leading power factor requirement at the high side of station transformer.

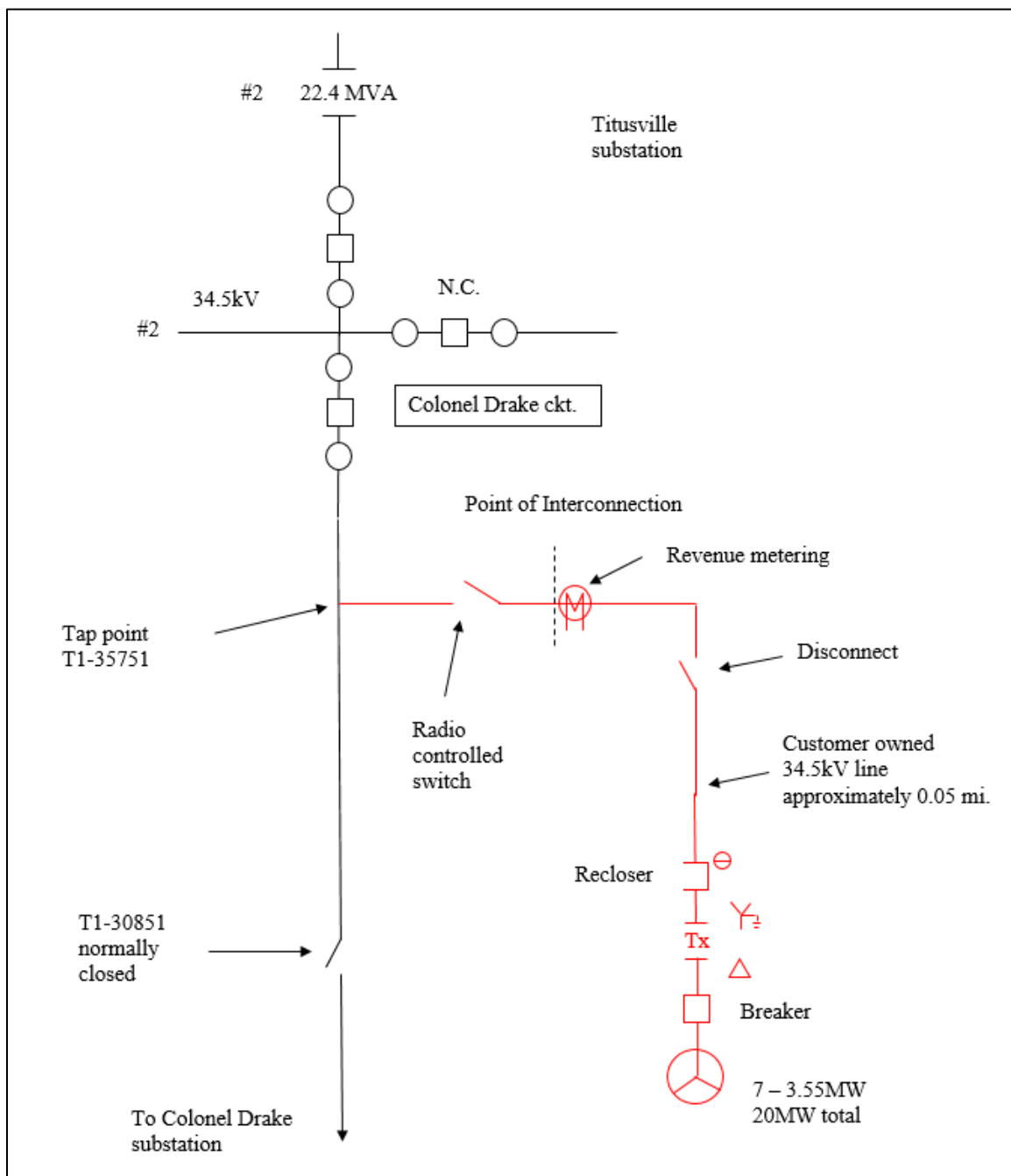
No mitigations are required.

Light Load

20 Light Load Analysis

Not applicable for solar projects.

21 Attachment 1- One Line Diagram



22 Attachment 2 – Project Location

